

1. Deep Learning Models for Video Processing.

1.1. *Convolutional Neural Networks for Image Classification.* Proposed by [Waibel et al. \(1989\)](#), *Convolutional Neural Network* (CNN) is a typical neural network for processing images. In 2012 Microsoft Image Classification Competition, a special variant of CNN named AlexNet [Krizhevsky, Sutskever and Hinton \(2012\)](#) won the championship of the competition and exceeded the classification accuracy of traditional machine learning methods by 15%, making people realize the great potential of CNN models in image processing. Inspired by the great success of AlexNet, more and more CNNs with various architectures were proposed in recent year, including VGG [Simonyan and Zisserman \(2014\)](#), GoogleNet [Szegedy et al. \(2015\)](#), ResNet [He et al. \(2016\)](#), Swin-Transformer [Liu et al. \(2021\)](#).

The good performance of CNNs mainly come from the following two points. First, a CNN utilizes the complex structure of neural networks and solves the problem of insufficient fitting in traditional machine learning models by providing a large number of parameters in image classification. Secondly, compared to traditional fully connected neural networks, the CNN model uses exquisite convolutional kernel design to effectively extract local features of the image, preventing overfitting caused by too many parameters.

1.2. *Region-Based Convolutional Neural Networks for Object Detection.* Object detection refers to the image processing problem of highlighting locations of interest objects in images by horizontal rectangles exactly including the target objects. Each highlighted rectangle is usually called a *Region of Interest* (RoI).

Proposed by [Girshick et al. \(2014\)](#), *Region-based Convolutional Neural Network* (R-CNN) is an important extension of CNN designed for object detection problem. R-CNN incorporates deep learning into object detection problem and overcomes the two basic limitations of traditional methods by automatically training the selection function of candidate regions and the corresponding region features via CNN. It consists of five main stages: a proposal stage that generates candidate regions for additional investigation; a feature-extraction stage that extracts important features for each proposed candidate region; a judgement stage that determines whether a candidate region contains an object of interest based on its feature vector; an adjustment stage that enhances the location and size of a candidate region; and a suppression stage that eliminates highly overlapping regions. Later on, many extensions of RNN have been proposed, including *Faster R-CNN* [He et al. \(2015\)](#), *YOLO* [Redmon et al. \(2016\)](#), *RetinaNet* [Lin et al. \(2017\)](#), *EfficientDet* [Tan, Pang and Le \(2020\)](#), *DiffusionDet* [Chen et al. \(2023\)](#).

1.3. *RNN, LSTM and attention mechanism for processing sequence data.* *Recurrent Neural Networks* (RNNs) refer to artificial neural networks characterized by a consistent architecture and layer flow. RNNs have been widely used for solving problems involving time series or sequence data, e.g., handwriting recognition, speech recognition, machine translation, robot control and so on, and gradually became the default choice for sequence data analysis in machine learning.

Proposed by [Hochreiter and Schmidhuber \(1997\)](#), *Long Short-Term Memory* (LSTM) is a variant of traditional RNN that can effectively capture the semantic association between long sequences and alleviate the phenomenon of gradient disappearance or explosion, with gate structures, including forget gate, input gate, cell state, and output gate. LSTM can better handle task involving time series or sequence data, because it solves the long-term dependency problem of RNN, and alleviates the "gradient disappearance" problem caused by back propagation in training RNN.

Although LSTM is an effective tool to process sequence data, it is inefficient to model long-distance temporal correlation in a data sequence. The *attention mechanism*, however, is

more powerful to handle long-distance temporal correlation by identifying the most relevant parts of the input sequences via a 2-dimensional attention map Vaswani et al. (2017). One of the most popular applications of attention mechanism is in the transformer model, which has been widely used in natural language processing and image processing.

1.4. *Existing methods for surgical video analysis.* Figure 1 shows the architecture of ToolNet, PhaseNet, EndoNet, SwinNet and some of their extensions equipped with LSTM and attention mechanism.

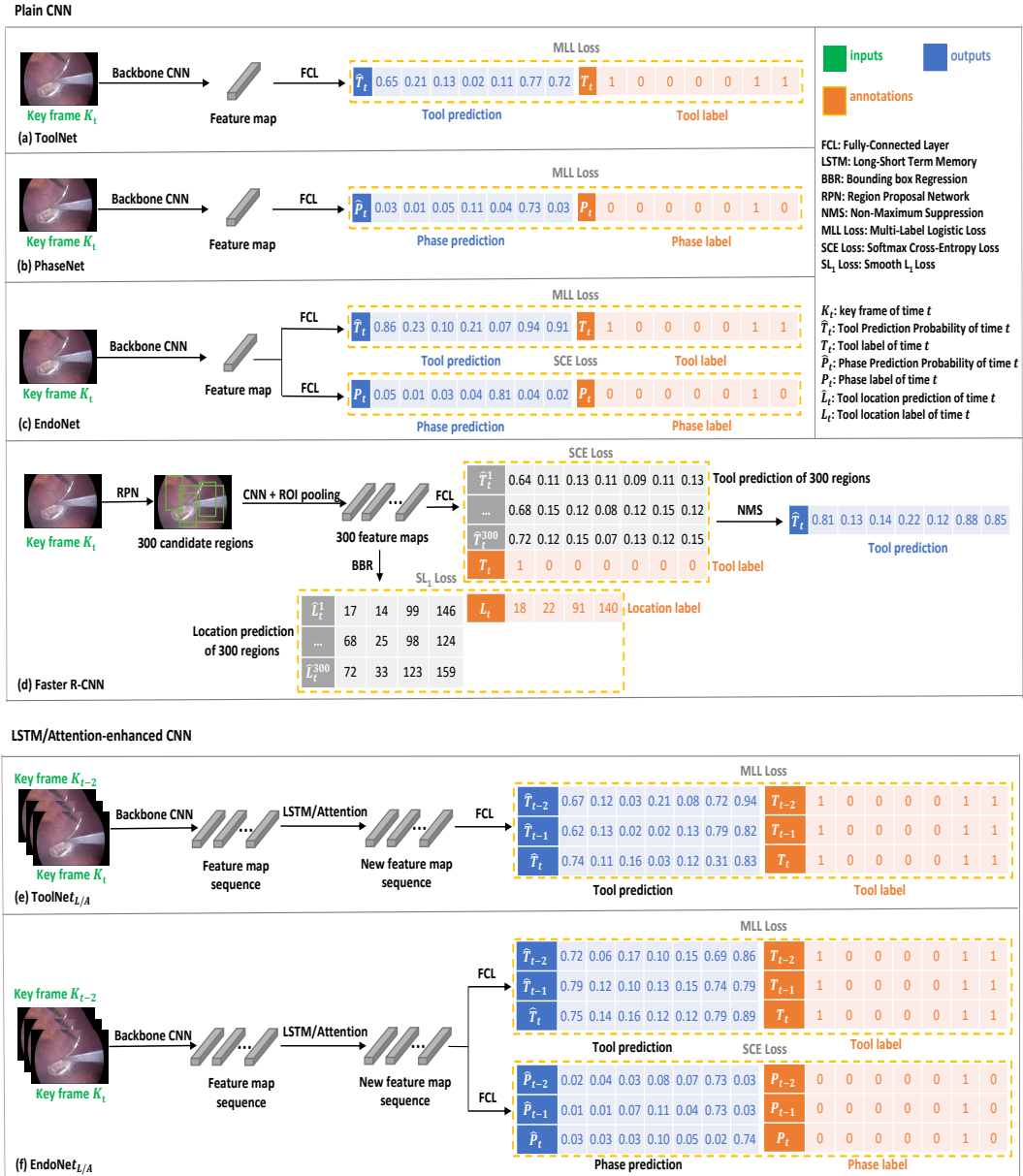


FIG 1. Graphical illustration of deep-learning methods for surgical tool recognition.

1.5. *Performance of MobileNet and MambaVision in Tool Presence Detection on Cholec80.* The results show that the proposed HMM-stabilized deep learning (HSDL) works well for MobileNet and Mambavision too, with an increase of mAP from 84.2 to 89.6 and 87.3 to 90.8 for the cholec80 dataset.

	MobileNet	MobileNet _H	MambaVision	MambaVision _H
Grasper	84.9	86.0	86.8	87.0
Bipolar	84.2	90.9	89.0	95.7
Hook	98.2	98.3	98.2	98.3
Scissors	73.8	79.8	76.0	76.6
Clipper	85.9	95.5	91.2	96.5
Irrigator	71.9	82.4	78.0	86.8
Speciman Bag	90.2	94.2	91.7	94.3
mAP	84.2	89.6	87.3	90.8

TABLE 1

Performance of MobileNet and MambaVision in Tool Presence Detection on Cholec80

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